European Journal of Political Economy 88 (2025) 102696

Contents lists available at ScienceDirect

European Journal of Political Economy

journal homepage: www.elsevier.com/locate/ejpe

Full Length Article

Spend on what? Insights on military spending efficiency

Dimitrios Dimitriou [a,b], Eleftherios Goulas [c], Christos Kallandranis [a,*]

[a] Department of Accounting and Finance, University of West Attica, Athens, Greece
[b] Department of Economics, National and Kapodistrian University of Athens, 1 Sofokleous str., Athens, Greece
[c] Department of Economics, University of Patras, University - Campus Rio, 26504, Patras, Greece

ARTICLE INFO

JEL classification:
O47
H56
O30
D74

Keywords:
Economic growth
Military spending
Efficiency
Defence alliance

ABSTRACT

This study investigates the role of military sector efficiency on the economic growth-defence spending relationship in 135 countries from 1992 to 2020. Efficiency stems from capital intensity of the military sector and the existence of technological externalities and economies of scale associated with armaments exports. Our findings indicate that a higher level of efficiency can moderate or eliminate the adverse effects of military burden on economic growth. Introducing alliance participation, we highlight the 'protective' role of NATO for its member-states with lower efficiency levels, while SCO seems to be less supportive.

1. Introduction

The debate over the influence of military spending on economic development, known as the 'guns or butter' dilemma, remains unresolved despite over five decades of inquiry. Increasing defense expenditures might result in a decline in financing for other socially important areas that are deemed more significant by the general population. Though, the present geopolitical circumstances require ongoing allocations of resources towards defense, both in the near and distant future. This entails prioritizing investments in novel technologies and R&D to effectively counter emerging threats and adapt to changing geopolitical dynamics.

The vast literature has identified various channels through which military spending can influence economic growth with diverse though impact (Alptekin and Levine, 2012; Emmanouilidis and Karpetis, 2020). Defence spending hinders economic growth by crowding out investment and consumption (e.g., Dunne and Tian, 2016; D'Agostino et al., 2017). On the other hand, defence expenditure can be growth-enhancing due to positive externalities stemming from technology spin-offs, human capital investment, and security spillovers (Awaworyi Churchill and Yew, 2018). Thus, the main argument is whether any losses arising from crowding out of resources devoted to the military sector exceed any positive externalities. This leads the net effect of defence burden[1] to be ambiguous

* Corresponding author.
E-mail addresses: ddimitriou@uniwa.gr (D. Dimitriou), egoulas@upatras.gr (E. Goulas), chriskal@uniwa.gr (C. Kallandranis).

[1] Military of defence burden is a crucial metric in defence economics, illustrating the trade-offs between military expenditure and other economic objectives with a social impact (Dunne and Tian, 2015). It is typically measured as the percentage of military spending to GDP.

https://doi.org/10.1016/j.ejpoleco.2025.102696
Received 23 November 2024; Received in revised form 7 April 2025; Accepted 13 May 2025
Available online 17 May 2025
0176-2680/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

since each channel conveys opposing effects. However, according to a study by Dunne and Tian (2016), the longer time series of data and the new panel data techniques that have become available lately, have shifted researchers to adopt the idea that the debate no longer emphatically holds as post-cold war data show in most cases a statistically significant negative effect of military spending on growth.[2] To avoid this shortcoming, it is rather obvious then that an alternative framework that disentangles these effects is necessary. The need for analyses that decompose military spending effects on growth has been put forward, with a focus on the efficiency as a key axis for judging this relationship (Kentor and Kick, 2008; Jorgenson et al., 2010). In fact, defense spending and efficiency[3] involve balancing the allocation of resources to ensure the security of a nation while also ensuring these resources are used effectively (Kettl, 2006).

In this respect, we move beyond traditional theories which concentrate on total defence spending, towards more complex explanations relating to the individual components making up such expenditures with a special focus first on military expenditures per soldier (MEPS) as it was introduced by Kentor (2004) and second on the ability of a country to produce and trade military equipment highlighting the economic significance of the military sector. Using a panel of 135 countries (see Appendix 1) during 1992–2020, we examine whether the response of economic growth on military spending depends on the degree of capital intensiveness[4] of the army, and the capacity of the country to export arms (technology externalities). It is expected that countries with more capital-intensive armies will convert more easily each additional amount spent in the military into growth-enhancing. By the same token, countries with the ability to export arms and military material, gain in terms of output growth due to spillover effects and economies of scale (see Blum, 2019). Both dimensions of military spending efficiency are considered either separately or as a pair. To the best of our knowledge, this is the first attempt to measure and compare the effects of both efficiency metrics. Our analysis goes beyond the usual approaches in the literature that mainly focus on the one or the other measure (see for example, Yakovlev, 2007). Next, we introduce the role of alliance membership, and we examine the differential effect of these indicators which might imitate the long-standing debate on burden sharing especially within NATO (Blum, 2019). To do this, we assume whether a country is a member of a defence alliance and approach the level of alliance integration by considering country-membership to vary between different types of alliances such as NATO (North Atlantic Treaty Organization) and SCO (Shanghai Cooperation Organization).

Our findings indicate that in countries with more capital-intensive armies, military burden[5] can be growth-enhancing compared to countries with less capital available. Further, countries with the ability to export military material and equipment can take advantage of any spillover effects and economies of scale, thus achieving to reverse the detrimental effects of military burden on growth. We further postulate that NATO is supportive in transforming military spending into growth-enhancing irrespective of the efficiency level of each member-state. On the contrary, access to the SCO alliance does not seem to be as 'protective' as NATO since its member-states with lower levels of efficiency experience a higher reduction in output growth due to the military burden they incur. This reflects the maturity level of each alliance in the form of network effects and other diffusion channels arising from the structure and operation of each organization.

The remainder of the paper is structured as follows: Section 2 provides a brief review of the prior literature on the role of efficiency on the economic growth-military spending relationship. Section 3 presents the empirical model and the identification strategy. Section 4 discusses the estimation results, while Section 5 concludes.

2. Military spending and efficiency effects

The investigation of disaggregated military expenditure aligns with a request from several scholars to shift the emphasis from total military spending to more nuanced analyses that account for the individual components of these expenditures (Kentor and Kick, 2008; Kentor et al., 2012). We firstly shift our attention to a specific component of military spending, which we designate as military spending per soldier. This serves as an indicator of the level of capital investment in a country's military infrastructure. The primary contention is that a move from a less capital-intensive army that relies heavily on human labour to one that depends more on capital will have a substantial impact on the amount of human expertise needed to effectively utilize the newly acquired capital. This expertise can be acquired primarily through education and technical training, leading to increased productivity. Indeed, capital-intensive armies prioritize advanced technology, armaments, and systems like drones and cyber defence mechanisms to enhance effectiveness, enhance individual warriors' abilities, and reduce reliance on large unit deployments (Biddle, 2006, 2017). The widespread use of unmanned aerial vehicles (UAVs) in surveillance and combat operations illustrates how capital-intensive investments can mitigate human danger while preserving operational efficacy (Singer, 2009). Such advanced military technology, frequently created with significant public

[2] Dunne and Tian (2013) report that from 168 studies published since Benoit's seminal work appeared in 1973, only 20 percent of studies found positive results of military spendings on economic growth. These results were also confirmed in their 2016 study for an extended sample. On the other, Alptekin and Levine (2012) found a positive effect in a meta-analysis using data from 1960 to 1990. However, with the addition of newer studies the positive association, is no longer valid (Awaworyi Churchill and Yew, 2018).

[3] Military efficiency denotes the ability of a military organization to proficiently transform its available resources into desired military results (Keupp, 2021).

[4] A capital-intensive army denotes a military force that depends significantly on advanced technology, specialised weaponry, and improved infrastructure instead of a substantial number of soldiers. This approach differs from labour-intensive armies, which rely mostly on the quantity of their military men over technological superiority, (Biddle, 2006).

[5] This concept provides an indication of how much of a country's economic output is dedicated to defense and security needs, reflecting the prioritization of military spending within the national economy (SIRPI Fact Sheet, April 2024).

funding, can stimulate innovation and technological spillovers into civilian industries, thus fostering overall economic growth (Mowery and Rosenberg, 1998). According to Kentor and Kick (2008), such capital intensity of military expenditures reflects the quality and structure of this form of spending, crucial in a world with significant technological advancements. Indeed, Kentor et al. (2012) indicate that high-tech expenditures on average doubled across a sample of 83 nations from 1970 to 1999, although military personnel experienced a minor reduction.

Modern warfare relies on electronics, communication, and information technology, with personnel adapting to these changes. Hence, capital-intensive militaries require more educated, but fewer, personnel. This necessitates, initially, a heightened level of intellect and knowledge for several individuals being admitted, beyond the previous requirements. This specifically excludes individuals who, in previous periods, relied on the military as a final option for employment. These high-tech expenditures have the greatest impact on the economies of less developed countries (Kentor and Jorgenson, 2017). Kentor and Kick (2008) argue that investments in advanced technology hinder a nation's progress by limiting workforce growth, restricting unskilled and unemployed individuals, and reducing the pool of funds to be invested in more economically beneficial opportunities. In this line, Jeofferey and Edward (2008) advocate that military expenditure deters per capita GDP growth, especially in less developed countries, due to the deceleration of labour force expansion. In the same vein, Kentor et al. (2012) support that the allocation of significant financial resources towards advanced military technology, as indicated by the ratio of military expenditures per soldier, worsens the inequality in household income, reducing current employment opportunities for less-qualified people.

It is rather obvious, that since the empirical relationship between military spending and growth is somewhat inconclusive, this connection can be seen under the prism of a different component of defence spending, that of trade and especially the arms trade. Such an interplay has been extensively studied within the context of military alliances, emphasising the interaction between security collaboration and economic interdependence. Military alliances frequently enhance trade among member states by mitigating the risks linked to geopolitical instability and cultivating trust between nations (Callado Muñoz et al., 2023a). This issue becomes even more apparent as NATO's requirement for a 2 % GDP contribution to defence expenditure puts at the centre stage the trade-offs associated with military alliances (Li et al., 2024). The decrease in transaction costs and the alignment of strategic objectives promote stronger economic connections, as alliances can provide a structure for conflict resolution and stability (Haim, 2016). Although the beneficial effects of military alliances on trade seem obvious, empirical studies provide once again mixed results (Li et al., 2024). Indeed, studies that present contrary evidence (e.g. Morrow et al., 1998; Mityakov et al., 2013 *etc.*) reveal that the impact of military alliances on trade is rather small or minimal. All in all, the degree to which military alliances promote trade is contingent upon the economic complementarities and historical relationships among member states. Certain researchers contend that alliances can redirect commerce by favouring intra-alliance partners over foreign entities (Davis and Meunier, 2011).

In this study, we are specifically exploring whether international trade in armaments, can act as an amplification mechanism on countries' industrial base and thus affecting economic growth (Mazzucato, 2011). It comes with no surprise that the largest contributors to military expenditures globally are also among the largest participants in the arms trade (Yakovlev, 2007). Following Rahman and Siddiqui (2019), global arms trade is a highly divisive topic. Countries import weapons for national security and economic reasons, while from the perspective of supply, governments engage in the sale of weapons for economic reasons as no country can independently produce all its own weapons (Brauer, 2007).

Within the context of the military expenditures and growth, the interaction of defence spending and arms trade dimension cannot be ignored as the former may affect growth through the channel of arms exports. If arms exports can help transferring a portion of the tax burden linked to military R&D financing, then it is reasonable to anticipate a direct link between economic growth and net exports of arms (Yakovlev, 2004). Military spending in the presence of arms exports can secure economic growth according to Rahman and Siddiqui (2019). This might explain at least for the affluent economies the controversial results in the literature so far, when the relationship between growth and defence spending is used under a linear specification. Indeed, while defence spending may divert funds from other productive activities and impose a tax burden on a country's citizens, it can also generate externalities and stimulate growth incentives. One may also argue that the dominance of developed or technologically advanced nations in the arms trade market indicates a significant level of technological advancement in these nations. Therefore, developed countries may encounter more significant technology benefits or spill-over effects resulting from increased military expenditure, which can be represented by arms exports. Gilli and Gilli (2016) argue that creating advanced weapon systems requires significant financial investment and expertise in design, development, and manufacture. This investment would benefit the defence manufacturing sector expanding its exports, and in turn would lead to economies of scale, increased employment, and technological advancements, ultimately boosting economic growth. Callado Muñoz et al. (2023b), provide evidence that shocks arising from military exports have an obvious spillover effect to labour productivity and real GDP per capita growth for the US. Besides, as arms development is increasingly focused on R&D, armaments exports are anticipated to have a beneficial effect on civil manufacture and innovation (Mazzucato, 2011). Indeed, according to Mowery (2010), the US government's defence purchases have been shown to stimulate innovation in both defence and civilian sectors. This expansion of public defence R&D has improved productivity and employment growth.

3. Methodology

3.1. Baseline model

The growth effects of military spending are examined using alternative approaches. The first assumes a Solow-style model following the empirical specification given in Knight et al. (1996). The economic growth equation originates from the Mankiw et al. (1992) version of the model of Solow (1956) and Swan (1956). Output growth is determined by initial output and a set of other

variables namely investment rate, human capital, population growth, international trade openness, conflict variables, and military spending. In this context, the Solow growth model provides a framework to analyze how military spending can influence economic growth through its effects on capital accumulation and technological progress. For instance, military spending can reduce the resources available for productive investment in physical and human capital. If financed by higher taxes or borrowing, it may crowd out private investment, leading to lower capital accumulation and long-run growth (Knight et al., 1996; Barro & Sala-i-Martin, 2004). In the same spirit, excessive spending can lead to inefficiencies and resource misallocation, reducing overall productivity and growth (Deger and Smith, 1983). On the contrary, military R&D can lead to technological advancements that spill over into civilian industries, boosting productivity (Ramey, 2011). Thus, while defense-related technological spillovers can boost growth, excessive military spending can crowd out private investment, potentially reducing long-term economic growth. Other studies that have adopted similar specifications include Murdoch and Sandler (2002), Dunne et al. (2005), Yakovlev (2007), Hou and Chen (2013) among others. The growth model employed here corresponds to the following specification:

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma\ln(milex)_{it} + \theta_i + n_t + \varepsilon_{it} \tag{1a}$$

where

$$X_{it,\kappa} = [\ln(sav)_{it}, \ln(hc)_{it}, \ln(popg)_{it}, \ln(open)_{it}, (coint)_{it}, (coend)_{it}]$$

Subscripts i and t refer to country and time, α, β_k 's and γ are unknown parameters to be estimated, θ_i and n_t are country- and time-fixed effects respectively, and ε_{it} is an error term. The dependent variable, $(growth)_{it}$, is real per-capita output growth. The lagged value of the logarithm of real GDP per capita, $ln(gdppc)_{it-1}$, will attain a negative coefficient if conditional convergence[6] applies. The variable of interest is the current value of the logarithm of military expenditure (burden) as percentage of GDP, $\ln(milex)_{it}$. Several studies conclude that any possible positive direct effects of defence spending on growth would arise through demand stimulation and other spin-off effects. On the contrary, any negative indirect effects would be the result of reductions in investment. However, Smith (2000) concluded that although the literature is unable to confirm any positive or negative or insignificant effect, there is some evidence in favour of a small negative impact in the long run. The vector of controls, $X_{it,\kappa}$, consists of standard determinants of economic growth, namely saving rate, $(sav)_{it}$, human capital, $(hc)_{it}$, population growth,[7] $(popg)_{it}$, trade openness, $(open)_{it}$, and two conflict-related dummies capturing the intensive of a conflict, $(coint)_{it}$, and the end of a conflict, $(coend)_{it}$. In particular, the saving rate proxied by the investment to GDP ratio, human capital measured by the number of years in secondary education, trade openness defined as the sum of exports and imports as a percentage of GDP, and the end of a conflict dummy are expected to exhibit a positive coefficient. On the other hand, population growth rate and the intensity of a conflict are assumed to have a decreasing effect on economic growth. All independent variables, except $(confint)_{it}$ and $(confend)_{it}$ which are defined as dummy variables, are expressed in the natural log form. We further include time dummies to control for world-wide growth of technology, as well as for other common shocks across countries that might have taken place during the period under consideration. Detailed variable definitions and data sources are presented in Appendix 2.

We further proceed by analyzing the growth effects of military spending utilizing a reformulated Barro growth model (Aizenman and Glick, 2006; Yakovlev, 2007). The reformulated model is an extension of Barro's (1990) endogenous growth model in that it highlights the **dual effects** of defense spending—potentially boosting growth through security and technology, but also hindering it if excessive taxation or crowding-out effects dominate. The reformulation aims to improve the original framework by addressing limitations such as the treatment of government spending, tax policies, and their effects on long-term economic growth. Further, the model expands upon the original by incorporating new factors, such as technological spillovers and institutional quality.

The Barro-style growth regression estimated here is of the following form:

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma(milex)_{it} + \theta_i + n_t + \varepsilon_{it} \tag{1b}$$

where

$$X_{it,\kappa} = \left[(sav)_{it}, ln(hc)_{it}, (popg)_{it}, (open)_{it}, (corr)_{it}, (gov)_{it}, (coint)_{it}, (coend)_{it}\right]$$

The set of control variables X includes the traditional growth variables[8] as described in (1a) as well as the level of corruption, $(corr)_{it}$, and the fraction of final government expenditure in GDP, $(gov)_{it}$. It is expected that increased taxation to finance more government consumption spending, introduces economic distortions, and therefore hinders economic growth (Barro, 1991; Brumm, 1997). In the same spirit, the level of corruption is assumed to deter economic growth as it reduces the efficiency of public spending (Aizenman and Glick, 2006; Pieroni and d'Agostino, 2009). This implies a negative coefficient on both $(gov)_{it}$, and $(corr)_{it}$. Note that only the number of years of schooling (human capital) is expressed in a log form following the Barro-style specification presented in Aizenman and Glick (2006) and Yakovlev (2007).

[6] It is expected that, other things equal, lower-income countries grow faster than higher-income countries.

[7] A common practice in Solow-style regressions is to calculate the log of population growth rate +0.05 (this is the assumed value for g + d). Following Mankiw et al. (1992), we assume technological progress plus the depreciation rate (g + d) to be the same for all countries and years and equal to 0.05.

[8] In the Barro-style growth regression, $(popg)_{it}$ is the annual population growth rate.

3.2. Measuring military spending efficiency

We examine potential heterogeneities in the economic growth-defence spending relationship in the presence of efficiency[9] arising from either the degree of capital intensity of the military sector, and/or the economies of scale and technological externalities stemming from the ability of the country to produce and export arms and high-tech military equipment. First, we proxy the quality of military forces by the level of military expenditure per soldier (MEPS) following the traditional view that military capability is a function of material resources. MEPS is a measure of the capital intensiveness[10] of the military sector as it reflects the amount of equipment or material relative to the size of the military personnel. Therefore, we aim to examine its direct and indirect impact on economic well-being. Second, we concentrate on alternative channels that are also crucial in transforming the military burden into growth-enhancing such as technology externalities and economies of scale. Especially in developed nations, the effectiveness of military expenses becomes higher as rich countries i) are more capable to producing equipment and other military material in a large-scale, ii) can operate more technologically advanced infrastructures, and iii) are endowed with high-skilled armed personnel (Beckley, 2010). All these aspects of efficiency are reflected in the ability of the country to produce technologically advanced military equipment and hence sell it abroad.

We classify a country i as having a higher-than-median degree of capital intensity when its annual military expenditure per soldier value, $(meps)_{it}$, is above the median value, $(meps)^{med}$, obtained from the distribution of countries with positive MEPS levels. Hence, the high MEPS dummy is defined as:

$$(hmeps)_{it} = \begin{cases} 1, if\ (meps)_{it} > (meps)^{med} \\ 0, otherwise \end{cases} \tag{2a}$$

Following Yakovlev (2007), we use net arms exports as a proxy for spill-over effects and economies of scale. We partition the sample between arms exporting and non-arms exporting nations by defining a dummy that equals one if the country i exhibits positive net arms exports (arms exports - arms imports) for a given year. Hence, the net arms exports dummy is defined as:

$$(armnx)_{it} = \begin{cases} 1, if\ (net\ arms\ exports)_{it} > 0 \\ 0, otherwise \end{cases} \tag{2b}$$

We amend the previous Solow-style model in (1a) to incorporate the interactions of defence expenditures with these influences:

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,k} + \gamma\ln(milex)_{it} + +\delta(hmeps)_{it} + \lambda(armnx)_{it} + \mu_h Z_{it,h} + \theta_i + n_t + \varepsilon_{it} \tag{3a}$$

where

$$Z_{it,h} = \left[(hmeps)_{it} \times \ln(milex)_{it}, (armnx)_{it} \times \ln(milex)_{it}\right]$$

Our priors are that countries with more available military capital stock per member of the military personnel will be able to utilize their resources more effectively compared to countries with limited military capital and equipment. Thus, capital intensity in the military sector is crucial in enhancing economic growth either directly or indirectly through its interaction with the size of the defence spending incurred. In other words, one would expect a positive coefficient δ for $(hmeps)_{it}$ and for its interaction with the logarithm of military spending to GDP ratio $(hmeps)_{it} \times \ln(milex)_{it}$. The degree of capital intensiveness is expected to play a crucial role in the formation of the growth-military spending relationship since it would be sufficient to reverse (or moderate) any harmful effects on economic growth.

Similarly, the existence of any technology spillovers stemming from military R&D and the ability of the country to produce military equipment on a large scale can be beneficial for economic growth as well. Undoubtedly, arms exporting countries tend to be more efficient and can adopt new technologies more easily in their production processes to improve competitiveness. We measure this by the capability of a country to export arms relative to the amount of its imports. In other words, we consider the ability of a country to not only being able to produce and thus export military equipment, but also its ability to export in such a scale that is adequate to override the amount of its arms imports, making therefore arms trade surplus. To be more specific, a country with limited quantity in its arms exports but with higher arms imports is considered as a non-arms exporting country because its arms trade balance is less than zero, though the level of its arms exports is positive. We anticipate a positive coefficient λ for $(armnx)_{it}$ and for its interaction with the logarithm of military spending to GDP ratio $(armnx)_{it} \times \ln(milex)_{it}$.

We also test whether the combination of both sources of efficiency produces an amplified effect given that countries with higher military capital stock is expected to exhibit greater capability to export arms. This enables us to test whether these efficiencies,

[9] Biddle (2017) designates the characteristics of numerical power, advancements in technology, and force deployment techniques, as essential in determining a military's effectiveness.

[10] Brauer (2002) argues that military expenditure per soldier is misleading in measuring capital intensity since only the equipment part of total military expenditure should be accounted for (without any personnel and indirect support expenses). Further, a stock variable should be used to measure capability instead of a flow variable. However, due to data unavailability on (disaggregated) military equipment for all the countries included in the sample, we resort to the MEPS measure for which series can be easily obtained for a large number of developed and less developed countries. This indicator of capital intensity has been adopted as a reliable and robust measure by many researchers or has been considered at least as the best available measure despite its weaknesses.

although arising from various sources, exhibit a complementary bahaviour in the formation of the economic growth and military spending relationship. We do this by introducing a triple interaction term i.e. $(hmeps)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$ that isolates this effect.

The analysis is not solely based on a Solow-style model. We further proceed by using the reformulated Barro-style model presented in (1b) and extend it to combine these interactions. The transformed model is presented below, where the set of control variables X has been amended to include the level of corruption, $(corr)_{it}$, and the fraction of final government expenditure in GDP, $(gov)_{it}$ as previously:

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma(milex)_{it} + +\delta(hmeps)_{it} + \lambda(armnx)_{it} + \mu_h Z_{it,h} + \theta_i + n_t + \varepsilon_{it} \tag{3b}$$

where

$$Z_{it,h} = \left[(hmeps)_{it} \times (milex)_{it}, (armnx)_{it} \times (milex)_{it}\right]$$

To address potential mismeasurements issues using MEPS, we employ arms trade per soldier (ATPS) as a proxy for capital intensiveness of the military sector. This indicator is calculated as the sum of arms imports and exports over military personnel. It is expected that arms imports represent a more accurate indicator of capital intensity of the army since they comprise mainly of military weapons (see Appendix 2 for a detailed description), which is a major part of the material used by the armed forces. Further, arms imports operate in a similar way as MEPS do, in that i) it is essential that military personnel will be trained and will obtain new skills to sufficiently use high-tech weapons systems, and ii) importing countries may be able to get new technology through reverse engineering (Yakovlev, 2007). We also add the value of arms exports to account for the capability of the country to domestically produce its own material. Although arms exports are military weapons imported by other countries, they represent domestic production that potentially could be counted as a part of the domestic military capital. Thus, we introduce ATPS as an alternative measure of the degree of capital intensiveness of the military sector and we test its validity in relation to the MEPS. However, arms imports and arms exports have limitations in measuring military capital. Countries with advanced defense industries such as the U.S. and Russia may produce most of their own weapons, making arms imports a poor measure of their military strength. Moreover, importing weapons and materials does not guarantee their sustained use, which seems to be influenced by other factors such as maintenance and training. Similarly, arms exports are often influenced by diplomatic relationships, sanctions, geopolitical considerations etc. Additionally, although arms exports are produced domestically, they are not formally considered as a part of the domestic military capital. Thus, we classify a country i as having a higher-than-median degree of capital intensity when its annual arms trade per soldier value, $(atps)_{it}$, is above the median value, $(atps)^{med}$, obtained from the distribution of countries with positive ATPS levels.

3.3. NATO vs. SCO membership

We assess the role of a country membership in the major defence alliances,[11] namely the North Atlantic Treaty Organization (NATO) and the Shanghai Cooperation Organization (SCO). To do so, we construct dummy variables for military alliance participation, which take the value of one if the country under consideration is a member of NATO or SCO in a given year. Military alliances influence global security and economic frameworks. A military alliance requires a strong political relationship between nations. Typically, countries with stronger political ties engage in increased trade, whereas political disputes frequently result in diminished economic exchanges (Jackson and Shepotylo, 2024). NATO and the SCO significantly vary in their origins, organisational frameworks, and strategic aims. The economic implications of military expenditures vary based on alliance-specific objectives in which member states' function. Our point of view is to analyze the economic consequences of military expenditures associated with NATO and SCO participation, focusing on how membership can influence resource distribution, economic development, and overall integration within an ally structure. To be more precise, NATO's 2022 Strategic Concept[12] focuses on its dedication to collective security and requires significant defence expenditures to counter hybrid threats, and ensure energy security (NATO, 2022). On the other, the SCO lacks an explicit collective defence agreement but prioritizes intelligence-sharing and collaborative counterterrorism exercises through its Regional Anti-Terrorist Structure (RATS). Despite their different strategic objectives, both NATO and SCO require significant investments in military technologies, infrastructure, and preparedness activities.

Our view is to examine whether NATO or SCO membership status of a country alters the direction, and the strength of the efficiency effects discussed in the previous section by allowing them to vary between groups of countries within the same alliance, as well as across types of alliances. Instead of assuming participation in any defence alliance, we proceed by distinguishing between specific alliance types. This will help us to account for the level of 'maturity' of each alliance and the strength of ties among allies.

The assumption here is that NATO membership benefits its member states through various mechanisms including the development of strong linkages between NATO allies, the assistance provided to its members to expand their military capabilities, and the availability of coordination mechanisms that focus on capability development and engagement with the defence industry. The notion that military equipment and technologies are more likely to be exchanged among allies with visible results in their economic growth is in line with previous research (Pamp et al., 2018; Callado Muñoz et al., 2023a). More specifically, NATO enables countries to obtain weapons systems, vehicles and other major equipment, and guarantees that national systems are able to integrate. Additionally, NATO supports collaboration with the defence industry on innovation, takes steps to ensure a sustained level of defence industrial production, and enhances cooperation across member-states in the form of common exercises and activities that lead to NATO collective

[11] See Dimitriou et al. (2024) for a brief discussion.
[12] See https://www.nato.int/strategic-concept/.

effectiveness. Other mechanisms include technology transfers, access to NATO common funds, mutual support, strategic mobility etc. All the above characteristics are of major importance, especially for allies with less developed military sectors that do not possess them in abundance. Indeed, according to Palaios et al. (2023), the U.S. and the North-European NATO members (the U.K., the Netherlands, and Germany) are generally net contributors to the alliance, whereas the South-European member states (Italy, Greece, and Turkey) are typically net receivers something that might corresponds to a free-riding behaviour (George and Sandler, 2018). On the other hand, the SCO is less integrated than NATO. Hence, any network effects are possibly weaker, indicating that more time is required to achieve a higher level of cohesion among its member-states. Thus, we construct a model that identifies how military spending efficiency interacts with defence alliance membership.

The models presented below in (6a)-(6d) differ from the model in (3a) in that the vector Z has been transformed to account for the alliance membership effects. We define a new vector Π that contains triple interaction terms between the log of military expenditure over GDP, the binary measure of efficiency, and the type of defence alliance the country participates in. Similarly, the vector T incorporates the simple interactions between each efficiency measure dummy and the defence alliance dummy respectively. A model specification of this form enables us to test the existence of any direct and indirect effects of these influences on economic growth. To facilitate our analysis, we define the dummy $(lmeps)_{it}$, which represents countries with a lower-than-median degree of capital intensity (i.e. $(meps)_{it} < (meps)^{med}$). Accordingly, the dummy $(non\ armnx)_{it}$ indicates non-arms exporters, and takes the value of one if the country i exhibits negative net arms exports for a given year. In the same spirit, the dummy variables $(non\ nato)_{it}$, $(non\ sco)_{it}$ take the value of one if the country is not a member of NATO or SCO for a given year. Therefore, we construct the following covariates to decompose the effect of military spending on growth across combinations of MEPS levels and alliance membership status:

$$T^{meps}_{it,nato} = [(nato)_{it} \times (hmeps)_{it}, (nato)_{it} \times (lmeps)_{it}, (non\ nato)_{it} \times (hmeps)_{it}] \tag{4a}$$

$$\Pi^{meps}_{it,nato} = \left[(nato)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}, (nato)_{it} \times (lmeps)_{it} \times \ln(milex)_{it},\ (non\ nato)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}\right] \tag{4b}$$

$$T^{meps}_{it,sco} = [(sco)_{it} \times (hmeps)_{it}, (sco)_{it} \times (lmeps)_{it}, (non\ sco)_{it} \times (hmeps)_{it}] \tag{4c}$$

$$\Pi^{meps}_{it,sco} = \left[(sco)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}, (sco)_{it} \times (lmeps)_{it} \times \ln(milex)_{it},\ (non\ sco)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}\right] \tag{4d}$$

We follow the same process for net arms exports to account for NATO and SCO membership accordingly:

$$T^{armnx}_{it,nato} = [(nato)_{it} \times (armnx)_{it}, (nato)_{it} \times (non\ armnx)_{it}, (non\ nato)_{it} \times (armnx)_{it}] \tag{5a}$$

$$\Pi^{armnx}_{it,nato} = [(nato)_{it} \times (armnx)_{it} \times \ln(milex)_{it}, (nato)_{it} \times (non\ armnx)_{it} \times \ln(milex)_{it},\ (non\ nato)_{it} \times (armnx)_{it} \times \ln(milex)_{it}] \tag{5b}$$

$$T^{armnx}_{it,sco} = [(sco)_{it} \times (armnx)_{it}, (sco)_{it} \times (non\ armnx)_{it}, (non\ sco)_{it} \times (armnx)_{it}] \tag{5c}$$

$$\Pi^{armnx}_{it,sco} = [(sco)_{it} \times (armnx)_{it} \times \ln(milex)_{it}, (sco)_{it} \times (non\ armnx)_{it} \times \ln(milex)_{it},\ (non\ sco)_{it} \times (armnx)_{it} \times \ln(milex)_{it}] \tag{5d}$$

Therefore, the Solow-type[13] model described in (3a) is amended as follows:

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma \ln(milex)_{it} + +\delta_h T^{meps}_{it,nato} + \mu_h \Pi^{meps}_{it,nato} + \theta_i + n_t + \varepsilon_{it} \tag{6a}$$

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma \ln(milex)_{it} + +\zeta_h T^{meps}_{it,sco} + \nu_h \Pi^{meps}_{it,sco} + \theta_i + n_t + \varepsilon_{it} \tag{6b}$$

and

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma \ln(milex)_{it} + +\delta_h T^{armnx}_{it,nato} + \mu_h \Pi^{armnx}_{it,nato} + \theta_i + n_t + \varepsilon_{it} \tag{6c}$$

$$(growth)_{it} = a\ln(gdppc)_{it-1} + \beta_k X_{it,\kappa} + \gamma \ln(milex)_{it} + +\zeta_h T^{armnx}_{it,sco} + \nu_h \Pi^{armnx}_{it,sco} + \theta_i + n_t + \varepsilon_{it} \tag{6d}$$

In (6a), a significant coefficient on the interaction terms of the vector Π will provide evidence for a differential response of economic growth to defence spending depending on whether i) the country is considered as a low- or high-MEPS and ii) the country is a member of NATO or not. Accordingly, in (6b) we test the differential effect of military spending on growth assuming SCO membership instead. In (6c)-(6d) we maintain the same framework of analysis; however, the efficiency measure now relies on the ability of the country to export arms, so we replace the previous measure of capital intensity used in (6a)-(6b). Specifically, (6c) distinguishes between arms exporting and non-arms exporting countries in conjunction with NATO participation or not, while (6d) replaces NATO with SCO membership. Furthermore, using the specifications from (6a)-(6d), we can estimate a model that nests the joint effects of both alliances. This will help us to test whether the coexistence of NATO and SCO alliances in a model harms the reliability of the previous specifications in terms of the effects attributed to each alliance and those distributed across the specific groups of allies.

[13] The Barro-type model described in (3b) can be transformed accordingly.

4. Empirical results

4.1. Baseline estimates and efficiency effects

We employ the fixed-effects estimation method to account for the existence of country-specific factors. Table 1 presents the descriptive statistics for the main variables included in the analysis, while Table 2 shows the estimation results of the Solow-style models. Column (1) in Table 2 gives the results of the baseline model in (1a), where the coefficient estimate on the log of military expenditure (% of GDP) is negative and significant at the 5 % level, indicating its harmful effect on economic growth. This result supports the argument that higher military expenditure takes away profitable investments in infrastructure and other economic sectors (e.g., Knight et al., 1996; Dunne and Nikolaidou, 2012). The coefficient estimate means that a 1 % increase in the fraction of military spending is associated with a decrease by 0.009 % in the annual GDP per capita growth. The parameter indicating conditional convergence is highly significant and carries the expected negative sign. This is in line with the related literature on economic growth that poor countries grow faster than rich countries (e.g., Barro and Sala-i-Martin, 1990; Sala-i-Martin, 1994). The coefficients on the log of saving rate and the log of population growth are significant at the 1 % level and their signs are in accordance with our priors (e.g., Dunne, 2012; Dunne and Tian, 2015; Becker and Dunne, 2023). Human capital (as measured by the number of schooling years) and trade openness convey positive and statistically significant coefficients, which agree with other studies in this field (e.g., Knight et al., 1996; Yamarik, 2011). Finally, the dummy representing conflict intensiveness appears with a negative coefficient, whereas the end of a conflict dummy attains a positive coefficient as expected (e.g., Murdoch and Sandler, 2002; Dunne and Tian, 2015).

We turn our attention to the model that encompasses the various efficiency measures and examine whether these might be able to reduce any detrimental effects on economic growth. Section 3.2 gives an analytical framework where the relevant assumptions associated with the model in (3a) are presented. Column (2) provides the estimation results assuming that capital intensity differs across country-groups by incorporating the term $(hmeps)_{it}$ and its interaction with the log of military expenditure (% of GDP), $(hmeps)_{it} \times \ln(milex)_{it}$. Thus, apart from any direct effects of MEPS on growth *per se*, we test whether there are additional effects associated with the level of military spending. It is obvious that the interaction term carries a positive and significant coefficient at the 5 % level and equals 0.008. The military burden for low-MEPS countries (i.e. countries with low capital intensity) is given by the coefficient on $\ln(milex)_{it}$ which is negative (=-0.016) and significant at the 1 % level. It is evident that in this group of countries an increase in military spending leads to lower economic growth. In other words, budget allocation in military sectors with low capital endowment is not effective, as it creates inefficiencies which erode the returns of each additional amount spent. On the other hand, in high-MEPS countries, defence spending is less harmful to economic growth with a total impact of −0.008 (i.e. the sum −0.016 + 0.008). At the 5 % level, we do not reject the null hypothesis (p-value = 0.06) that this sum is equal to zero (H_0: $\gamma+\mu_{meps}=0$), leading thus to a marginally insignificant coefficient. More capital availability translates into more resources available, which indicates a less negative total effect on economic growth. This finding remains robust even after controlling for the direct effect of $(hmeps)_{it}$ on economic growth, which is positive and significant at the 5 % level.

Column (3) displays the results obtained when the effectiveness of military expenditures is measured by net arms exports as proposed by Yakovlev (2007). This indicator reflects aspects of economic efficiency that cannot be captured by capital intensity such as i) the ability to operate in a large scale and benefit from economies of scale, and ii) the gains obtained by the presence of technology externalities stemming from military R&D and other research-oriented activities. These features, common to all countries with the ability to produce and export military equipment and material, are measured by the binary variable $(armnx)_{it}$, which is constructed to splitting the sample between arms exporting and non-arms exporting countries, based on whether the value of net arms exports is greater than zero or not. We find that military burden is heavier for non-arms exporting countries (i.e. for countries with non-positive net arms exports) since the coefficient on $\ln(milex)_{it}$ is negative (=-0.010) and significant at the 1 % level. On the contrary, military spending for the arms exporting countries becomes no detrimental for growth as the corresponding estimate turns positive with a value

Table 1
Summary statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
$(growth)_{it}$	3120	0.022	0.039	−0.226	0.189
$(gdppc)_{it}$	3120	13.128	17.866	0.364	105.455
$(sav)_{it}$	3071	0.227	0.065	0.030	0.935
$(popg)_{it}$	3120	0.013	0.012	−0.038	0.076
$(hc)_{it}$	3120	6.438	0.920	4.000	9.000
$(open)_{it}$	3118	0.843	0.526	0.147	4.373
$(corr)_{it}$	2753	0.468	0.161	0.060	0.750
$(gov)_{it}$	3108	0.158	0.055	0.009	0.588
$(milex)_{it}$	3120	0.020	0.016	0.000	0.143
$(meps)_{it}$	2875	0.051	0.083	0.000	1.153
$(atps)_{it}$	2875	0.002	0.005	0.000	0.119
$(net\ arms\ exports)_{it}$	3120	0.040	1.147	−5.366	15.448
$(coint)_{it}$	3120	0.032	0.175	0.000	1.000
$(coend)_{it}$	3120	0.045	0.206	0.000	1.000

Notes: *Net arms exports* are expressed in US$ B, *meps* and *atps* are measured in US$ M per soldier, while *gdppc* denotes US$ K.

Table 2
Defence spending and effectiveness indicators in the Solow growth model.

VARIABLES	Baseline Model	MEPS	Net Arms Exports	MEPS & Net Arms Exports	MEPS with Net Arms Exports
	(1)	(2)	(3)	(4)	(5)
$\ln(milex)_{it}$	−0.009**	−0.016***	−0.010***	−0.017***	−0.013***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
$(hmeps)_{it}$		0.043**		0.040**	
		(0.016)		(0.016)	
$(hmeps)_{it} \times \ln(milex)_{it}$		0.008**		0.007*	
		(0.004)		(0.004)	
$(armnx)_{it}$			0.041**	0.042**	
			(0.017)	(0.018)	
$(armnx)_{it} \times \ln(milex)_{it}$			0.011**	0.011**	
			(0.004)	(0.004)	
$(hmeps)_{it} \times (armnx)_{it}$					0.074***
					(0.023)
$(hmeps)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$					0.018***
					(0.005)
$\ln(gdppc)_{it-1}$	−0.039***	−0.047***	−0.040***	−0.048***	−0.044***
	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)
$\ln(sav)_{it}$	0.035***	0.036***	0.035***	0.036***	0.037***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\ln(popg)_{it}$	−0.051***	−0.056***	−0.050***	−0.055***	−0.054***
	(0.011)	(0.013)	(0.011)	(0.013)	(0.013)
$\ln(hc)_{it}$	0.040***	0.046***	0.040***	0.047***	0.049***
	(0.014)	(0.015)	(0.014)	(0.015)	(0.014)
$\ln(open)_{it}$	0.016***	0.017***	0.017***	0.018***	0.017***
	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)
$(coint)_{it}$	−0.009***	−0.008***	−0.009***	−0.008***	−0.009***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
$(coend)_{it}$	0.004	0.006**	0.005	0.006**	0.007**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Constant	0.146**	0.157**	0.153**	0.164**	0.148**
	(0.063)	(0.070)	(0.064)	(0.071)	(0.071)
Observations	2866	2633	2866	2633	2633
R-squared	0.469	0.435	0.471	0.437	0.432
H_0: $\gamma+\mu_{meps}=0$		0.06		0.03	
H_0: $\gamma+\mu_{armnx}=0$			0.91	0.27	
H_0: $\gamma+\mu_{meps*armnx}=0$					0.37
H_0: $\mu_{meps}=\mu_{armnx}$				0.50	

Notes: Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. Country and time dummies are included in all specifications.

of 0.001 (the sum −0.010 + 0.011) and becomes insignificant (see hypothesis testing).

In column (4) we present the results assuming the inclusion of both measures of military effectiveness simultaneously. This will enable us to get more precise estimates of the contribution of each measure as the effect of MEPS is estimated in the presence of net arms exports and vice versa. Comparing the results in column (4) with that displayed in columns (2) & (3), we observe that the coefficients on both interaction terms remain statistically significant, with similar coefficient magnitudes and the expected positive signs. We do not reject the null hypothesis that the relevant coefficients are equal to each other i.e. H_0: $\mu_{meps}=\mu_{armnx}$. This provides evidence that the effects of both efficiency measures are statistically equal in the formation of the growth effect of military spending. Finally, in column (5) we evaluate their combined effect by identifying countries that are characterized by both a higher level of military capital and a higher ability to produce and export military equipment. The estimate on the triple interaction term is highly significant at the 1 % level indicating a positive total effect of military spending on growth equal to 0.005 (the sum −0.013 + 0.018), though insignificant.

To evaluate the sensitivity of our results across various specifications of the growth model, we display in Table 3 the estimates of (1b) & (3b) where a reformulated Barro-type model is employed. In columns (1)–(2) we present the baseline Barro model. The first specification is similar to the Solow model presented in column (1) of Table 2. The only difference is that only lagged GDP per capita and human capital appear in logs. The second specification is an augmented version where government final consumption and the corruption index have been added. In both columns, military burden is negative and statistically significant. Incorporating military capital effectiveness in columns (3)–(4), we document a positive and significant coefficient at the 5 % level for each interaction term $(hmeps)_{it} \times (milex)_{it}$ irrespective of the model specification. Similar conclusions are drawn when net arms exports are considered as a source of military efficiency in columns (5)–(6), and when both efficiency measures are simultaneously considered in columns (7)–(8). Finally, the pair of regressions in columns (9)–(10) reveal that when both sources coexist, the harmful effect of military spending on growth either disappears or turns positive.

Table 3
Defence spending and effectiveness indicators in the Barro growth model.

VARIABLES	Baseline Model		MEPS		Net Arms Exports		MEPS & Net Arms Exports		MEPS with Net Arms Exports	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$(milex)_{it}$	−0.424***	−0.297*	−0.906***	−0.672***	−0.459***	−0.325**	−0.937***	−0.700***	−0.541***	−0.295*
	(0.150)	(0.155)	(0.247)	(0.239)	(0.153)	(0.158)	(0.248)	(0.239)	(0.159)	(0.157)
$(hmeps)_{it}$			−0.001	−0.000			−0.001	0.000		
			(0.005)	(0.005)			(0.005)	(0.005)		
$(hmeps)_{it} \times (milex)_{it}$			0.563**	0.551**			0.557**	0.544**		
			(0.216)	(0.216)			(0.214)	(0.215)		
$(armnx)_{it}$					−0.011***	−0.010**	−0.012***	−0.010**		
					(0.004)	(0.004)	(0.004)	(0.004)		
$(armnx)_{it} \times (milex)_{it}$					0.418**	0.300*	0.426**	0.346**		
					(0.173)	(0.174)	(0.171)	(0.174)		
$(hmeps)_{it} \times (armnx)_{it}$									−0.011**	−0.007*
									(0.005)	(0.004)
$(hmeps)_{it} \times (armnx)_{it} \times (milex)_{it}$									0.542**	0.417**
									(0.218)	(0.187)
$\ln(gdppc)_{it-1}$	−0.039***	−0.051***	−0.047***	−0.055***	−0.040***	−0.051***	−0.048***	−0.056***	−0.043***	−0.051***
	(0.005)	(0.006)	(0.006)	(0.006)	(0.005)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$(sav)_{it}$	0.146***	0.159***	0.152***	0.158***	0.145***	0.158***	0.151***	0.158***	0.154***	0.161***
	(0.031)	(0.034)	(0.034)	(0.035)	(0.031)	(0.034)	(0.034)	(0.035)	(0.034)	(0.035)
$(popg)_{it}$	−0.936***	−1.222***	−1.022***	−1.235***	−0.936***	−1.222***	−1.023***	−1.235***	−1.016***	−1.224***
	(0.153)	(0.174)	(0.180)	(0.177)	(0.150)	(0.170)	(0.176)	(0.173)	(0.179)	(0.176)
$\ln(hc)_{it}$	0.044***	0.059***	0.049***	0.057***	0.044***	0.059***	0.049***	0.057***	0.052***	0.061***
	(0.015)	(0.018)	(0.016)	(0.019)	(0.015)	(0.018)	(0.017)	(0.019)	(0.015)	(0.018)
$(open)_{it}$	0.021***	0.016***	0.024***	0.022***	0.021***	0.017***	0.024***	0.022***	0.022***	0.020***
	(0.005)	(0.005)	(0.006)	(0.006)	(0.005)	(0.005)	(0.006)	(0.006)	(0.005)	(0.005)
$(coint)_{it}$	−0.009***	−0.010***	−0.008**	−0.008***	−0.008***	−0.010***	−0.008**	−0.008***	−0.009***	−0.010***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
$(coend)_{it}$	0.004	0.003	0.006**	0.005*	0.004	0.003	0.006**	0.005*	0.006**	0.005**
	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
$(corr)_{it}$		−0.078***		−0.073**		−0.078***		−0.073***		−0.068**
		(0.029)		(0.028)		(0.028)		(0.028)		(0.027)
$(gov)_{it}$		−0.173***		−0.200***		−0.170***		−0.196***		−0.200***
		(0.060)		(0.056)		(0.060)		(0.056)		(0.057)
Constant	0.233***	0.367***	0.289***	0.407***	0.239***	0.372***	0.296***	0.412***	0.251***	0.359***
	(0.044)	(0.060)	(0.051)	(0.062)	(0.045)	(0.061)	(0.051)	(0.062)	(0.050)	(0.060)
Observations	2886	2545	2651	2503	2886	2545	2651	2503	2651	2503
R-squared	0.467	0.455	0.433	0.476	0.468	0.456	0.435	0.477	0.429	0.470

Notes: Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. Country and time dummies are included in all specifications.

4.2. Arellano-Bond GMM estimates

When analyzing the effect of military spending on economic growth one needs to take into account endogeneity, which arises because military spending may both influence and be influenced by economic growth. To handle the issue of simultaneity in a regression analysis we resort to the Arellano and Bond (1991) GMM estimator where lags of the endogenous variable are used as instruments. This approach ensures that only past values influence the current dependent variable (Becker and Dunne, 2023). A similar econometric approach has been followed by other researchers including Caselli et al. (1996) and Hou and Chen (2013) among others. In Table 4 we display the estimations of the Solow-style models from (1a) & (3a). In the Arellano-Bond GMM, the first differences of the growth equation are calculated to remove the unobserved country-specific effects, and then in this first-differenced equation, lagged levels of the RHS variables are used as instruments. Following the suggestions by Roodman (2009), we restrict the number of instruments used by collapsing the instrument set. This enables us to avoid problems associated with instrument proliferation such as overfitting endogenous variables. Further, we also add time-specific effects to reduce the influence of cross-sectional error dependence (Ding and Knight, 2011).

Specification tests are applied to check the validity of the instruments in our GMM estimation. The AR(1) test checks for first-order serial correlation in the residuals of the differenced equation, while the AR(2) test evaluates whether the residuals exhibit second-order serial correlation. This is more important for assessing the validity of the instruments because the presence of second-order correlation in the differenced residuals would imply that the instruments are endogenous or invalid. Failing to reject the null hypothesis of no second-order serial correlation provides evidence that the instruments are valid and that the model is correctly specified. In all columns of Table 4, the hypothesis of no second-order serial correlation is not rejected at any level of significance. Finally, the Hansen test is applied to test the null hypothesis of instrument validity. Failure to reject this null hypothesis means that the instruments are valid. The Hansen statistic confirms the joint validity of the instruments used (i.e., the instruments are not correlated with the error term), implying that all models are well specified. Clustered robust standard errors at the country level are assumed in all regressions.

The results displayed in Table 4 confirm the findings using the fixed-effects method in Table 2. The estimates on the corresponding interaction terms with each measure of efficiency remain positive and consistently significant but with much higher magnitudes, indicating that any individual source of efficiency is adequate to fully reverse the harmful effects of defence burden on growth. The same is also evident when the efficiency measures appear simultaneously in column (4). We are not able to reject the null hypothesis that the sums $\gamma+\mu_{meps}$ and $\gamma+\mu_{armnx}$ are equal to zero, revealing a non-significant impact of defence burden on economic growth for high-MEPS or arms-exporting countries. Again, we confirm that both sources statistically produce an equal moderating impact as we do not reject the null hypothesis that the relevant coefficients are equal to each other (see H_0: $\mu_{meps}=\mu_{armnx}$). In column (5) we observe that their combined effect is stronger since the corresponding interaction term exhibits a statistically significant coefficient which is larger in size. In particular, its size is 4.5 times higher (in absolute terms) than that on $\ln(milex)_{it}$, thus leading to a total positive effect on economic growth which is equal to 0.073 and statistically significant at the 5 % level (see H_0: $\gamma+\mu_{meps*armnx}=0$).

4.3. Defence alliance membership: NATO vs SCO

At this point, we go a step further by bringing into our attention an additional source of heterogeneity stemming from the decision of a country to participate in a defence alliance. The study by Dimitriou et al. (2024) is one of the first attempts to examine the type of defence alliance membership in the formation of the growth-military spending relationship. Their analysis controls for the type of alliance by conducting comparative analysis across NATO and non-NATO states. In the same context, we consider NATO and SCO[14] participation by estimating (6a)-(6d) in Table 5. This enables us to test whether the findings in Table 2 differ across different types of alliances, and between country-groups of the same alliance.

The coefficients on the control variables appear with the expected signs and significance in all regressions. However, their estimates are not reported due to space conservation. In column (1), the coefficients on the triple interaction terms $(nato)_{it} \times (lmeps)_{it} \times \ln(milex)_{it}$ and $(nato)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}$ are positive and highly significant. Both are of equal size, which is also confirmed by statistical testing (p-value = 0.87). This finding indicates that network effects due to NATO membership could be more helpful for member-states that lack military capital. In other words, NATO is an integrated organization that offers more advantages to countries with more traditional military sectors (i.e. with less capital availability) by providing resources, sophistication, technology transfers, technical support etc. This process would enable less capital-endowed states to catch up with the most advanced military nations indicating therefore possible free-riding responses (Goerge and Sandler, 2024; Qari et al., 2024). On the other hand, the coefficient on the interaction term that refers to the non-NATO high-MEPS states is statistically insignificant and smaller in size (=0.006) indicating that NATO high-MEPS member-countries gain efficiency advantages compared to their non-NATO peers. Each amount spent on military purposes by NATO governments becomes more valuable, since NATO members can increase their capability frontier through collaboration with advanced nations, irrespective of whether they are considered as high-MEPS countries or not. Examining the direct effects of the pairs between NATO participation and capital intensity level, we document positive and statistically significant parameters for all cases, with NATO countries exhibiting the highest coefficients. In column (2) we provide the estimates when the impact of military spending is examined through the capability of a country to export arms in conjunction with NATO membership or not. We reveal a positive impact for NATO member-states, which is symmetric in statistical terms as we fail to reject the

[14] The SCO alliance has been selected as a comparable to NATO as it represents a regional multilateral organization, which is multifunctional, and has started dynamically to devote itself mainly to security goals showing a remarkable speed of evolution.

Table 4
Arellano-Bond GMM estimates on defence spending and efficiency indicators.

VARIABLES	Baseline Model	MEPS	Net Arms Exports	MEPS & Net Arms Exports	MEPS with Net Arms Exports
	(1)	(2)	(3)	(4)	(5)
$\ln(milex)_{it}$	−0.014**	−0.034***	−0.022***	−0.038***	−0.021***
	(0.006)	(0.010)	(0.007)	(0.010)	(0.007)
$(hmeps)_{it}$		0.226**		0.202*	
		(0.105)		(0.103)	
$(hmeps)_{it} \times \ln(milex)_{it}$		0.051*		0.045*	
		(0.027)		(0.026)	
$(armnx)_{it}$			0.246**	0.229**	
			(0.114)	(0.096)	
$(armnx)_{it} \times \ln(milex)_{it}$			0.067**	0.061***	
			(0.027)	(0.023)	
$(hmeps)_{it} \times (armnx)_{it}$					0.355**
					(0.155)
$(hmeps)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$					0.094**
					(0.037)
$\ln(gdppc)_{it-1}$	−0.048**	−0.072**	−0.055**	−0.079***	−0.046*
	(0.024)	(0.029)	(0.025)	(0.028)	(0.026)
$\ln(sav)_{it}$	0.080***	0.088***	0.083***	0.089***	0.094***
	(0.019)	(0.022)	(0.019)	(0.022)	(0.022)
$\ln(popg)_{it}$	−0.119***	−0.109***	−0.121***	−0.106***	−0.098***
	(0.033)	(0.038)	(0.033)	(0.036)	(0.036)
$\ln(hc)_{it}$	0.049*	0.039	0.041	0.032	0.038
	(0.028)	(0.026)	(0.027)	(0.026)	(0.027)
$\ln(open)_{it}$	0.035	0.048*	0.014	0.040	0.034
	(0.028)	(0.029)	(0.030)	(0.029)	(0.030)
$(coint)_{it}$	−0.010**	−0.010*	−0.009*	−0.009	−0.007
	(0.004)	(0.006)	(0.005)	(0.006)	(0.005)
$(coend)_{it}$	0.007**	0.009***	0.007**	0.010***	0.009***
	(0.003)	(0.004)	(0.003)	(0.004)	(0.003)
Observations	2692	2451	2692	2451	2451
AR(1), p-value	0.00	0.00	0.00	0.00	0.00
AR(2), p-value	0.25	0.56	0.39	0.63	0.58
Hansen, p-value	0.67	0.55	0.75	0.64	0.59
Number of Instruments	158	149	160	151	149
H_0: $\gamma+\mu_{meps}=0$		0.41		0.77	
H_0: $\gamma+\mu_{armnx}=0$			0.08	0.32	
H_0: $\gamma+\mu_{meps*armnx}=0$					0.04
H_0: $\mu_{meps}=\mu_{armnx}$				0.63	

Notes: Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. AR(1) and AR(2) are residual first and second order serial correlation tests, while Hansen stands for the over-identifying restrictions test. Country and time dummies are included in all specifications.

null of equal coefficients (p-value = 0.22), although in quantitative terms the NATO arms-exporting countries gain more. The effect for the non-NATO countries with higher ability to export military equipment is smaller (=0.009) and marginally significant, indicating that NATO collaboration raises the efficient use of military resources in a country.

Columns (3)–(4) display the estimates when participation in the SCO alliance is considered. Column (3) provides the estimates based on the level of capital intensiveness, which is measured by MEPS. It is evident that for the SCO member-states, defence spending is detrimental for growth, however the negative effect is accentuated for low-MEPS countries. Further, we reject the null that the effects between high- and low-MEPS countries of the SCO alliance are equally distributed (p-value = 0.00). In column (4) we test whether defence burden is harmful for growth by considering net arms exports and SCO participation as well. It is apparent that any unfavourable effects on growth become more visible for the SCO countries with no ability to accelerate growth through technology externalities and economies of scale. The corresponding coefficient is negative, marginally significant and almost 10 times higher (in absolute terms) than that on the interaction term indicating the SCO arms exporting countries. In the group of non-SCO countries (including also NATO members) with positive arms external balance, economic growth is profited by spill-over effects as the corresponding interaction term conveys a positive and highly significant coefficient.

Finally, in columns (5)–(6) we proceed by estimating the model incorporating NATO and SCO membership simultaneously. The terms denoting non-NATO and non-SCO countries have been removed from the new specification. The model in this form concentrates on the effects of defence spending on growth between country-groups belonging solely to NATO or the SCO alliances, therefore making easier for us to disentangle these effects from the non-alliance countries. In both columns, similar conclusions can be drawn as previously. It is evident that between NATO country-groups the asymmetry is not revealed. Both country-groups are benefited in terms of additional growth gains. On the other hand, the asymmetric effects in the SCO case are more profound and consistent, confirming the heterogeneity between the SCO countries. Apart from the distribution of the effects among defence alliances we are also able to distinguish between alliance and non-alliance members. The coefficient on the variable $\ln(milex)_{it}$ measures the impact for the

Table 5
Defence spending and effectiveness indicators across NATO and SCO countries.

VARIABLES	NATO		SCO		NATO vs SCO	
	MEPS	Net Arms Exports	MEPS	Net Arms Exports	MEPS	Net Arms Exports
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(milex)_{it}$	−0.018***	−0.012***	−0.016***	−0.011***	−0.015***	−0.012***
	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
$(nato)_{it} \times (lmeps)_{it}$	0.126***				0.104**	
	(0.043)				(0.045)	
$(nato)_{it} \times (hmeps)_{it}$	0.133***				0.114***	
	(0.037)				(0.037)	
$(non\ nato)_{it} \times (hmeps)_{it}$	0.035*					
	(0.018)					
$(nato)_{it} \times (lmeps)_{it} \times \ln(milex)_{it}$	0.031***				0.026**	
	(0.010)				(0.010)	
$(nato)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}$	0.030***				0.027***	
	(0.008)				(0.008)	
$(non\ nato)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}$	0.006					
	(0.004)					
$(nato)_{it} \times (non\ armnx)_{it}$		0.096**				0.095**
		(0.042)				(0.042)
$(nato)_{it} \times (armnx)_{it}$		0.150***				0.146***
		(0.036)				(0.035)
$(non\ nato)_{it} \times (armnx)_{it}$		0.028				
		(0.019)				
$(nato)_{it} \times (non\ armnx)_{it} \times \ln(milex)_{it}$		0.024**				0.023**
		(0.010)				(0.010)
$(nato)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$		0.036***				0.035***
		(0.008)				(0.008)
$(non\ nato)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$		0.009*				
		(0.004)				
$(sco)_{it} \times (lmeps)_{it}$			−0.212**		−0.218**	
			(0.095)		(0.095)	
$(sco)_{it} \times (hmeps)_{it}$			−0.105		−0.116	
			(0.094)		(0.095)	
$(non\ sco)_{it} \times (hmeps)_{it}$			0.039**			
			(0.017)			
$(sco)_{it} \times (lmeps)_{it} \times \ln(milex)_{it}$			−0.067**		−0.068**	
			(0.033)		(0.033)	
$(sco)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}$			−0.044		−0.046	
			(0.031)		(0.031)	
$(non\ sco)_{it} \times (hmeps)_{it} \times \ln(milex)_{it}$			0.007*			
			(0.004)			
$(sco)_{it} \times (non\ armnx)_{it}$				−0.279*		−0.278*
				(0.166)		(0.167)
$(sco)_{it} \times (armnx)_{it}$				0.035		0.023
				(0.110)		(0.112)
$(non\ sco)_{it} \times (armnx)_{it}$				0.044**		
				(0.017)		
$(sco)_{it} \times (non\ armnx)_{it} \times \ln(milex)_{it}$				−0.084*		−0.084*
				(0.048)		(0.048)
$(sco)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$				−0.008		−0.010
				(0.036)		(0.036)
$(non\ sco)_{it} \times (armnx)_{it} \times \ln(milex)_{it}$				0.011***		
				(0.004)		
Constant	0.146**	0.150**	0.187***	0.178***	0.178**	0.166**
	(0.069)	(0.064)	(0.071)	(0.066)	(0.071)	(0.065)
Observations	2633	2866	2633	2866	2633	2866
R-squared	0.438	0.474	0.442	0.478	0.442	0.479
H_0: $\mu_{nato\ x\ lmeps}=\mu_{nato\ x\ hmeps}$	0.87				0.96	
H_0: $\nu_{sco\ x\ lmeps}=\nu_{sco\ x\ hmeps}$			0.00		0.00	
H_0: $\mu_{nato\ x\ non\text{-}armnx}=\mu_{nato\ x\ armnx}$		0.22				0.25
H_0: $\nu_{sco\ x\ non\text{-}armnx}=\nu_{sco\ x\ armnx}$				0.00		0.00

Notes: Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. The coefficients on control variables are not reported due to space limitations. Country and time dummies are included in all specifications.

remaining countries that are considered as non-NATO and non-SCO countries as well. This is always negative and statistically significant. Our results reflect that military spending for NATO countries is at least not detrimental to economic growth irrespective of the level of capital accumulation or the existence of technology spillovers in the military sector. On the other hand, military burden is clearly more detrimental for low-MEPS or non-arms exporting SCO countries.

4.4. ATPS as a measure of capital intensity

To assess the validity of our results we employ the arms trade per soldier (ATPS) as an alternative indicator of capital intensity of the military sector. Table 6 provides the estimates of the Solow growth model, while Section 3.2 provides a discussion about ATPS and its features.

The dummy variables $(hatps)_{it}$ and $(latps)_{it}$ measure high- and low-ATPS countries respectively. Focusing on the parameter of main interest, in column (1), the coefficient on the interaction term $(hatps)_{it} \times \ln(milex)_{it}$ is positive and significant at the 5 % level,

Table 6
Defence spending and ATPS across NATO and SCO countries.

VARIABLES	ATPS	NATO	SCO	NATO vs SCO
	(1)	(2)	(3)	(4)
$\ln(milex)_{it}$	−0.013***	−0.015***	−0.014***	−0.014***
	(0.004)	(0.004)	(0.004)	(0.004)
$(hatps)_{it}$	0.021**			
	(0.010)			
$(hatps)_{it} \times \ln(milex)_{it}$	0.005**			
	(0.002)			
$(nato)_{it} \times (latps)_{it}$		0.089**		0.085*
		(0.044)		(0.045)
$(nato)_{it} \times (hatps)_{it}$		0.133***		0.126***
		(0.036)		(0.037)
$(non\ nato)_{it} \times (hatps)_{it}$		0.016		
		(0.010)		
$(nato)_{it} \times (latps)_{it} \times \ln(milex)_{it}$		0.021**		0.020**
		(0.010)		(0.010)
$(nato)_{it} \times (hatps)_{it} \times \ln(milex)_{it}$		0.033***		0.031***
		(0.008)		(0.008)
$(non\ nato)_{it} \times (hatps)_{it} \times \ln(milex)_{it}$		0.003		
		(0.002)		
$(sco)_{it} \times (latps)_{it}$			−0.257*	−0.261*
			(0.139)	(0.139)
$(sco)_{it} \times (hatps)_{it}$			−0.061	−0.071
			(0.062)	(0.061)
$(non\ sco)_{it} \times (hatps)_{it}$			0.022**	
			(0.010)	
$(sco)_{it} \times (latps)_{it} \times \ln(milex)_{it}$			−0.079*	−0.080*
			(0.043)	(0.043)
$(sco)_{it} \times (hatps)_{it} \times \ln(milex)_{it}$			−0.036	−0.038
			(0.025)	(0.025)
$(non\ sco)_{it} \times (hatps)_{it} \times \ln(milex)_{it}$			0.005**	
			(0.002)	
$\ln(gdppc)_{it-1}$	−0.043***	−0.044***	−0.048***	−0.048***
	(0.006)	(0.006)	(0.006)	(0.006)
$\ln(sav)_{it}$	0.037***	0.036***	0.037***	0.036***
	(0.006)	(0.006)	(0.006)	(0.006)
$\ln(popg)_{it}$	−0.056***	−0.056***	−0.056***	−0.055***
	(0.013)	(0.013)	(0.013)	(0.012)
$\ln(hc)_{it}$	0.048***	0.047***	0.049***	0.048***
	(0.014)	(0.013)	(0.013)	(0.013)
$\ln(open)_{it}$	0.016***	0.018***	0.015***	0.017***
	(0.006)	(0.006)	(0.006)	(0.006)
$(coint)_{it}$	−0.010***	−0.011***	−0.010***	−0.010***
	(0.003)	(0.003)	(0.003)	(0.003)
$(coend)_{it}$	0.006**	0.007**	0.006**	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)
Constant	0.137*	0.136*	0.174**	0.175**
	(0.071)	(0.069)	(0.072)	(0.070)
Observations	2633	2633	2633	2633
R-squared	0.431	0.435	0.440	0.442
H_0: $\mu_{nato\ x\ latps}=\mu_{nato\ x\ hatps}$		0.12		0.14
H_0: $\nu_{sco\ x\ latps}=\nu_{sco\ x\ hatps}$			0.02	0.02

Notes: Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. Country and time dummies are included in all specifications.

advocating that military spending can be less damaging for growth in countries with ATPS levels above the median value. In columns (2)–(4) we introduce participation in a defence alliance. In the case of NATO, we report a positive and significant coefficient (at least at the 5 % level) for each of the interaction terms $(nato)_{it} \times (latps)_{it} \times \ln(milex)_{it}$ and $(nato)_{it} \times (hatps)_{it} \times \ln(milex)_{it}$, which is equal to 0.021 and 0.033 respectively. However, they are equal in statistical terms as the null hypothesis is not rejected (p-value = 0.12). Regarding the countries that are not members of NATO there is no significant impact of military spending, though it remains positive. In column (3) we consider SCO country membership. The estimated coefficients on the main interaction terms are both negative, however within the SCO group of countries, the adverse effect of military spending is statistically significant only for the country-group with the lower military capital availability (classified as low-ATPS countries). At the 5 % level, we also reject the null that both parameters are the same. Thus, we can conclude that more military capital is crucial in reducing any damaging effects on growth associated with military burden, however, it becomes more important to countries without strong NATO alliance networks. In the last column, similar conclusions are obtained when NATO and SCO membership is taken into account simultaneously.

5. Conclusion

The majority of research regarding the influence of military expenditure on economic growth is extensive and expanding, however, the consensus remains elusive. Using fixed-effects, and Arellano-Bond GMM estimators alternatively, we investigate for 135 countries from 1992 to 2020, whether the growth effects of each additional amount spent to the military is heterogeneous with respect to i) capital availability and the existence of technology externalities, and ii) the type of the defence alliance a country participates in, proxied by NATO and SCO membership. In countries with more capital-intensive armies, any additional amount spent becomes growth-enhancing compared to countries with less capital-intensive armies. Similarly, countries with the capacity to produce and export military material and equipment benefit by spillover effects and economies of scale thus creating a net gain in output growth. Our conclusions are mixed after considering defence alliance membership of a country and the type of alliance as well. It is documented by the empirical analysis that NATO membership is supportive in transforming military spending into growth-enhancing. On the contrary, economic growth in SCO member-states is not elevated by their access to the alliance. In this context, among NATO member-states there are no great differences in the effects of defence expenditure on growth even if we distinguish between NATO countries with more and less efficient military sectors. However, among the SCO member-states, the harmful effects of military burden on growth are amplified only for the group of countries with lower efficiency levels.

This study uncovers that the advantages of NATO membership outweigh those of standard military alliances. NATO reflects an international organization with a higher level of integration since its member-states can take advantage of the network effects that exist through its structure and operation, including military cooperation, knowledge externalities, common activities among allies etc. In the context of increased geopolitical tensions and evolving alliances, comprehending these processes is essential for both policymakers and researchers. International alliances are expected to significantly influence future military spending and economic growth patterns, as they often result in collective defense obligations, which can either increase or decrease national military expenditures. NATO members have encountered diverse effects on their defence spending as a result of collective security agreements. Subsequent studies in the relevant literature should possibly continue examining the economic aspects of military alliances, taking into account their immediate effects on trade, labour market innovation along with their wider ramifications for global security and growth. Indeed, the aspect of alliance dynamics can significantly affect global economic balances through trade relations, foreign direct investment, and geopolitical stability. Thus, understanding the future economic implications of military alliances is a discussion worth engaging by policymakers.

CRediT authorship contribution statement

Dimitrios Dimitriou: Writing – review & editing, Writing – original draft, Resources, Formal analysis, Conceptualization. **Eleftherios Goulas:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Christos Kallandranis:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejpoleco.2025.102696.

Data availability

Data will be made available on request.

References

Alptekin, A., Levine, P., 2012. Military expenditure and economic growth: a meta-analysis. Eur. J. Polit. Econ. 28, 636–650.
Aizenman, J., Glick, R., 2006. Military expenditure, threats, and growth. J. Int. Trade Econ. Dev. 15 (2), 129–155.
Arellano, M., Bond, S., 1991. Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. Rev. Econ. Stud. 58, 277–297.
Awaworyi Churchill, S., Yew, S.L., 2018. The effect of military expenditure on growth: an empirical synthesis. Empir. Econ. 55, 1357–1387.
Barro, R.J., 1990. Government spending in a simple model of endogenous growth. J. Polit. Econ. 98 (5), 103–125.
Barro, R.J., 1991. Economic growth in a cross section of countries. Q. J. Econ. 106 (2), 407–443.
Barro, R.J., Sala-i-Martin, X., 1990. Economic Growth and Convergence across the United States.
Barro, R.J., Sala-i-Martin, X., 2004. Economic Growth, second ed. MIT Press.
Becker, J., Dunne, J.P., 2023. Military spending composition and economic growth. Defence Peace Econ. 34 (3), 259–271.
Beckley, M., 2010. Economic development and military effectiveness. J. Strat. Stud. 33 (1), 43–79.
Biddle, S., 2006. Military Power: Explaining Victory and Defeat in Modern Battle. Princeton University Press.
Biddle, S., 2017. Military Effectiveness. Oxford Research Encyclopedia of International Studies.
Blum, J., 2019. Arms production, national defense spending and arms trade: examining supply and demand. Eur. J. Polit. Econ. 60, 101814.
Brauer, J., 2002. Survey and review of the defense economics literature on Greece and Turkey: what have we learned? Defence Peace Econ. 13 (2), 85–107.
Brauer, J., 2007. Arms industries, arms trade and developing countries. Handb. Defense Econ. 2, 973–1015.
Brumm, H.J., 1997. Military spending, government disarray, and economic growth: a cross-country empirical analysis. J. Macroecon. 19 (4), 827–838.
Callado Muñoz, F.J., Hromcová, J., Laborda Herrero, R., Utrero González, N., 2023a. An empirical analysis of arms exports and economic growth spillovers: the case of the United States. Defence Peace Econ. 34 (7), 893–913.
Callado Muñoz, F.J., Hromcová, J., Utrero González, N., 2023b. Can buying weapons from your friends make you better off? Evidence from NATO. Econ. Modell. 118, 106084.
Caselli, F., Esquivel, G., Lefort, F., 1996. Reopening the convergence debate: a new look at cross-country growth empirics. J. Econ. Growth 1, 363–389.
D'Agostino, G., Dunne, J.P., Pieroni, L., 2017. Does military spending matter for long-run growth? Defence Peace Econ. 28, 429–436.
Davis, C.L., Meunier, S., 2011. Business as usual? Economic responses to political tensions. Am. J. Polit. Sci. 55 (3), 628–646.
Deger, S., Smith, R., 1983. Military expenditure and growth in less developed countries. J. Conflict Resolut. 27 (2), 335–353.
Dimitriou, D., Goulas, E., Kallandranis, C., Drakos, K., 2024. Military expenditures and economic growth: evidence from NATO and non-NATO alliances. Defence Peace Econ. 1–25.
Ding, S., Knight, J., 2011. Why has China grown so fast? The role of physical and human capital formation. Oxf. Bull. Econ. Stat. 73 (2), 141–174.
Dunne, J.P., 2012. Military spending, growth, development and conflict. Defence Peace Econ. 23 (6), 549–557.
Dunne, J.P., Nikolaidou, E., 2012. Defence spending and economic growth in the EU 15. Defence Peace Econ. 23 (6), 537–548.
Dunne, J.P., Smith, R.P., Willenbockel, D., 2005. Models of military expenditure and growth: a critical review. Defence Peace Econ. 16, 449–461.
Dunne, J.P., Tian, N., 2013. Military expenditure and economic growth: a survey. Econ. Peace Secur. J. 8 (1), 5–11.
Dunne, J.P., Tian, N., 2015. Military expenditure, economic growth and heterogeneity. Defence Peace Econ. 26 (1), 15–31.
Dunne, J.P., Tian, N., 2016. Military expenditure and economic growth, 1960–2014. Econ Peace Securi. J. 11 (2).
Emmanouilidis, K., Karpetis, C., 2020. The defense–growth nexus: a review of time series methods and empirical results. Defence Peace Econ. 31, 86–104.
George, J., Sandler, T., 2018. Demand for military spending in NATO, 1968–2015: a spatial panel approach. Eur. J. Polit. Econ. 53, 222–236.
George, J., Sandler, T., 2024. A spatial analysis of NATO burden sharing at the operational levels. Kyklos 77 (4), 1026–1047.
Gilli, A., Gilli, M., 2016. The diffusion of drone warfare? Industrial, organizational, and infrastructural constraints. Secur. Stud. 25 (1), 50–84.
Haim, D., 2016. Alliance networks and trade: the effect of indirect political alliances on bilateral trade flows. J. Peace Res. 53 (3), 472–490.
Hou, N., Chen, B., 2013. Military expenditure and economic growth in developing countries: evidence from system GMM estimates. Defence Peace Econ. 24 (3), 183–193.
Jackson, K., Shepotylo, O., 2024. Political alliances and trade: Europe in a polarized world. Eur. J. Polit. Econ. (85), 102612
Jorgenson, A.K., Clark, B., Kentor, J., 2010. Militarization and the environment: a panel study of carbon dioxide emissions and the ecological footprints of nations, 1970-2000. Glob. Environ. Polit. 10 (1), 7–29.
Kentor, J., 2004. Quantifying hegemony in the world-economy. In: Reifer, T. (Ed.), Globalization, Hegemony & Power, Antisystemic Movements and the Global System. Paradigm Publishers.
Kentor, J., Jorgenson, A., 2017. Military expenditures and health: a cross-national study, 1975-2000. Int. J. Sociol. Soc. Pol. 37 (13/14), 755–772.
Kentor, J., Kick, E., 2008. Bringing the military back in: military expenditures and economic growth 1990 to 2003. J. World Syst. Res. 142–172.
Kentor, J., Jorgenson, A.K., Kick, E., 2012. The "new" military and income inequality: a cross-national analysis. Soc. Sci. Res. 41 (3), 514–526.
Kettl, D.F., 2006. Modernising government: the way forward-A comment. Int. Rev. Adm. Sci. 72 (3), 313–317.
Keupp, M.M., 2021. Efficiency of military performance. In: Defense Economics. Springer, Cham.
Knight, M., Loayza, N., Villanueva, D., 1996. The peace dividend: military spending cuts and economic growth. IMF Staff Pap. 43 (1), 1–37.
Li, B., Ge, Y., Lin, Y.-m., 2024. Military alliance and international trade: evidence from NATO enlargement. World Econ. 47, 930–956.
Mankiw, N.G., Romer, D., Weil, D.N., 1992. A contribution to the empirics of economic growth. Q. J. Econ. 107 (2), 407–437.
Mazzucato, M., 2011. The entrepreneurial state. Soundings 49 (49), 131–142.
Mityakov, S., Tang, H., Tsui, K., 2013. International politics and import diversification. J. Law Econ. 56 (4), 1091–1121.
Morrow, J., Siverson, R., Tabares, T., 1998. The political determinants of international trade: the major powers, 1907–90. Am. Polit. Sci. Rev. 92 (3), 649–661.
Mowery, D.C., Rosenberg, N., 1998. Paths of Innovation: Technological Change in 20th-Century America. Cambridge University Press.
Mowery, D.C., 2010. Military R&D and innovation. Handb. Econ. Innov. 2, 1219–1256. North-Holland.
Murdoch, J.C., Sandler, T., 2002. Economic growth, civil wars, and spatial spillovers. J. Conflict Resolut. 46 (1), 91–110.
Palaios, P., Triantafillou, A., Zombanakis, G., 2023. Net spills among NATO allies: theory and empirical evidence from dynamic quantile connectedness. Security and Defence Quarterly 2023 44 (4), 21–37.
Pamp, O., Dendorfer, F., Thurner, P.W., 2018. Arm your friends and save on defence? The impact of arms exports on military expenditures. Public Choice 177, 165–187.
Pieroni, L., d'Agostino, G., 2009. Military spending, corruption and economic growth. Peace Econ. Peace Sci. Publ. Pol. 14 (3).
Qari, S., Börger, T., Lohse, T., Meyerhoff, J., 2024. The value of national defense: assessing public preferences for defense policy options. Eur. J. Polit. Econ. 85, 102595.
Rahman, T., Siddiqui, D.A., 2019. The Effect of Military Spending on Economic Growth in the Presence of Arms Trade: a Global Analysis. Available at: SSRN 3401331.
Ramey, V.A., 2011. Identifying government spending shocks: it's all in the timing. Q. J. Econ. 126 (1), 1–50.
Roodman, D., 2009. How to do xtabond2: an introduction to difference and system GMM in Stata. STATA J. 9 (1), 86–136.
Sala-i-Martin, X., 1994. Cross-sectional regressions and the empirics of economic growth. Eur. Econ. Rev. 38 (3–4), 739–747.
Singer, P.W., 2009. Wired for War: the Robotics Revolution and Conflict in the 21st Century. Penguin Press.

Smith, R.P., 2000. Defence expenditure and economic growth. Making Peace Pay: A Bibliography on Disarmament and Conversion, pp. 15–24.
Solow, R., 1956. A contribution to the theory of economic growth. Q. J. Econ. 50, 65–94.
Swan, T., 1956. Economic growth and capital accumulation. Econ. Rec. 32, 334–361.
Yakovlev, P., 2004. Do arms exports stimulate economic growth? Defence Peace Econ. 18 (4), 317–338.
Yakovlev, P., 2007. Arms trade, military spending, and economic growth. Defence Peace Econ. 18 (4), 317–338.
Yamarik, S., 2011. Human capital and state-level economic growth: what is the contribution of schooling? Ann. Reg. Sci. 47, 195–211.